

Towards Triangulated Birepresentations: Evaluation and Induction in Soergel Categories

M. Mackaay V. Miemietz P. Vaz*

*Université catholique de Louvain

TSIMF Conference on Hopf Algebras and Tensor Categories

January 2026

What is this talk about?

- Representation theory of algebras studies homomorphisms

$$A \longrightarrow \text{End}(V).$$

- Representation theory of bicategories: actions on categories, i.e.

$$\mathcal{C} \longrightarrow \text{End}(\mathcal{M}).$$

- These are called **2-representations**.

This talk is about 2-representations coming from affine Hecke algebras.

The story in one picture

affine Hecke algebras $\longrightarrow$ Soergel categories
representations $\longrightarrow$ 2-representations
evaluation, induction $\longrightarrow$ categorified evaluation, induction

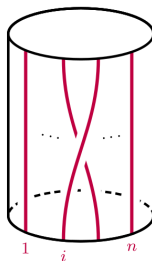
We start from affine Hecke algebras and their representations.

Why this problem?

- Finite-dimensional reps of affine Hecke algebras are rich and subtle.
- Two fundamental tools to study them:
 - evaluation representations,
 - parabolic (Zelevinsky) induction.
- Affine-type categorification leads to non-finitary settings:
 - infinitely many indecomposable 1-morphisms,
 - no finitary Hom-categories,
 - passing to homotopy / triangulated categories becomes unavoidable when looking at 2-representations. **This is a new phenomenon.**

Extended affine Hecke algebra $\widehat{H}_n^{\text{ext}}$:

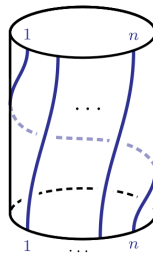
- Generators T_i and rotation $\rho^{\pm 1}$.
- Relations $\rho T_i \rho^{-1} = T_{i+1}$,
 $(T_i + q)(T_i - q^{-1}) = 0, \dots$
- $H_n \subset \widehat{H}_n^{\text{ext}}$ f.d. subalgebra generated by $T_1, \dots, T_{n-1}$.



T_i

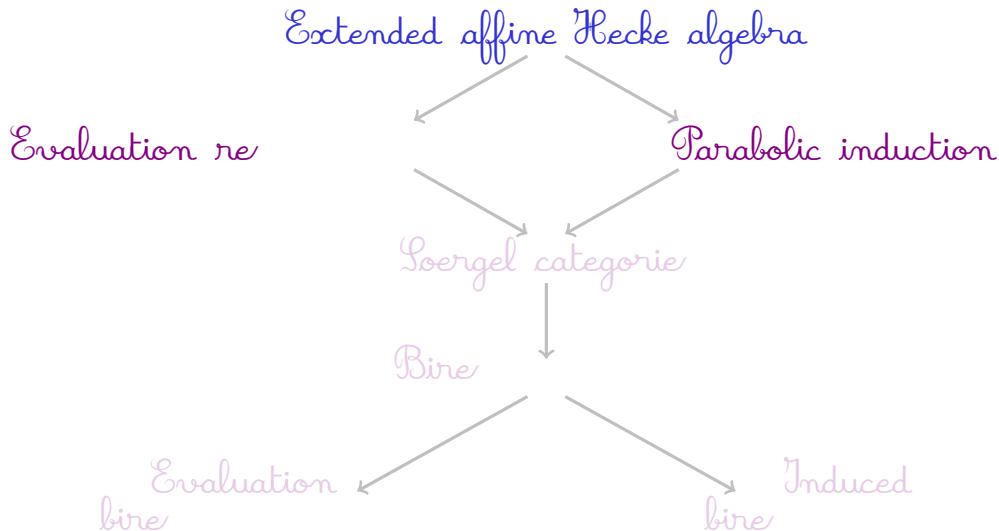
$(1 \leq i \leq n)$

indices mod n



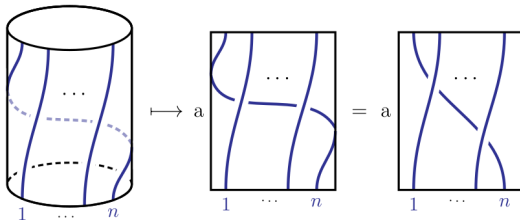
ρ

Two constructions of representations



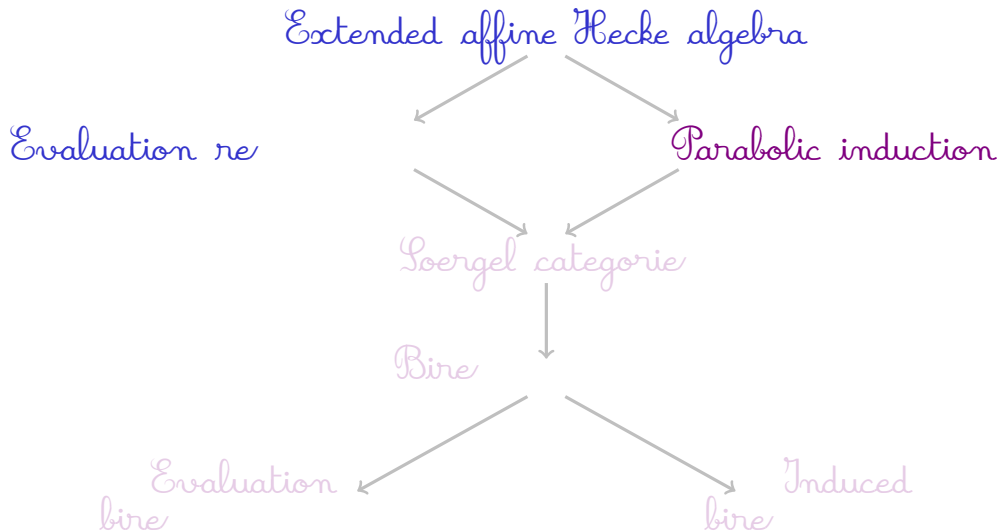
- For $a \in \mathbb{C}(q)^\times$: $\text{ev}_a : \widehat{H}_n^{\text{ext}} \longrightarrow H_n$

- Defined by
$$\begin{cases} T_i \mapsto T_i & (i \neq 0) \\ \rho \mapsto a T_{n-1} \cdots T_1 \end{cases}$$



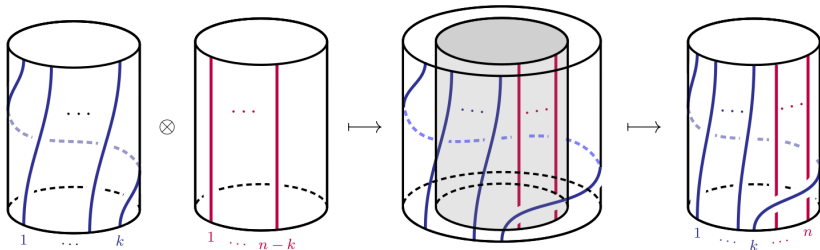
- Pullback produces evaluation representations.
- Key ingredient in finite-dimensional affine type A theory.

Two constructions of representations

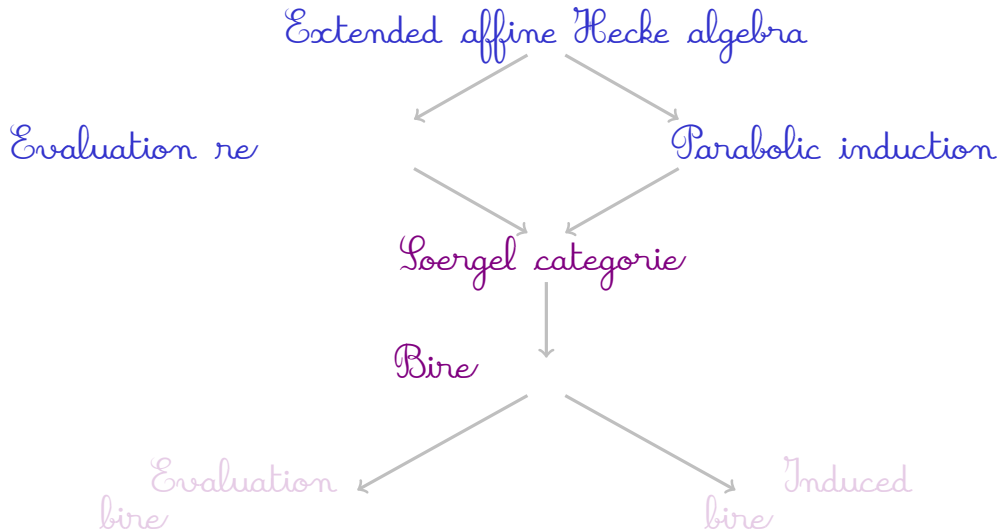


Classical parabolic induction (maximal parabolic)

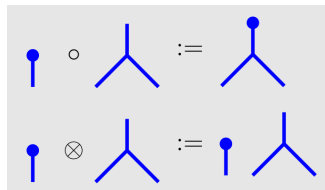
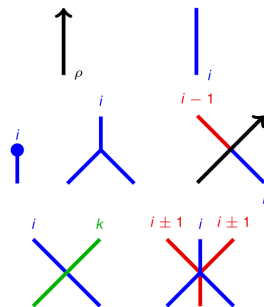
- Decomposition $n = k + (n - k)$ gives a maximal parabolic $S_k \times S_{n-k} \subset S_n$.
- Induction via algebra embeddings: $\hat{H}_k^{\text{ext}} \otimes \hat{H}_{n-k}^{\text{ext}} \hookrightarrow \hat{H}_n^{\text{ext}}$.



- Zelevinsky tensor product: $M_1 \odot M_2 := \text{Ind}(M_1 \otimes M_2)$.



- Soergel categories categorify Hecke algebras :
 - They are $\mathbb{K}$ -linear, additive, and monoidal (one-object bicategories).
 - $K_0(\mathcal{S}(W)) \cong H(W)$ (W Coxeter group).
 - Kazhdan–Lusztig generators $b_i := T_i + q$.
 - B_i categorifies b_i
- Extended affine type A category $\widehat{\mathcal{S}}_n^{\text{ext}}$.
- Diagrammatic calculus encodes relations.



Soergel cats categorify Hecke algebras, so they must act on categories.

- Classical representation theory studies algebra homomorphisms

$$A \longrightarrow \operatorname{End}(V), \quad (V \text{ a vector space})$$

- Categorify V to a category $\mathcal{M}$: replace $\operatorname{End}(V)$ by the bicategory $\operatorname{End}(\mathcal{M})$ of endofunctors and natural transformations.
- Formally, a *2-representation* of a bicategory $\mathcal{C}$ is a pseudofunctor

$$\mathcal{C} \longrightarrow \operatorname{End}(\mathcal{M}) \subset \operatorname{Cat}.$$

- In this talk, a *birepresentation* of a bicategory $\mathcal{C}$ means a (wide) finitary 2-representation in the sense of Mazorchuk–Miemietz and Macpherson.
- For Soergel categories we will always work in this 2-representation framework, and later in its triangulated analogue.

- *Finitary birepresentations* (Mazorchuk–Miemietz, 2011): 2-representations where all Hom-categories are equivalent to finitary additive categories.
 - Perfect for categorifying representations of finite-dimensional algebras.

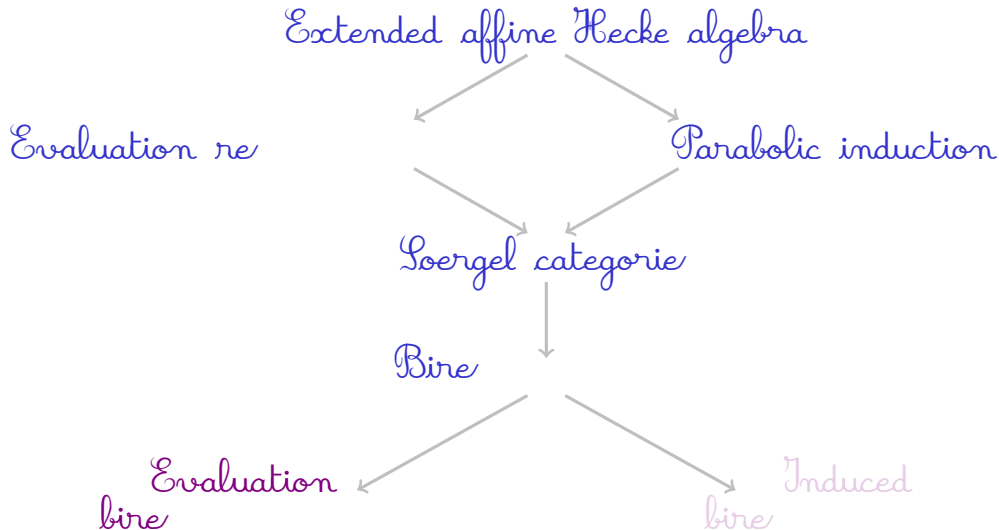
In affine type, Soergel categories have infinitely many indecomposable 1-morphisms, so **the finitary condition fails**.

- We therefore use Macpherson's *wide finitary 2-representations* (2022), which allow infinitely many indecomposable 1-morphisms while keeping each Hom-category finitary.

Why triangulated 2-representations?

- Evaluation and induction functors are built using Rouquier complexes.
- These live naturally in the homotopy category $K^b(\mathcal{S})$, not in the Soergel category $\mathcal{S}$ itself.

The correct framework for our 2-representations is **triangulated** rather than abelian or finitary



Goal: categorify ev_a .

- Monoidal functors: $\text{Ev}_{r,s}: \widehat{\mathcal{S}}_n^{\text{ext}} \rightarrow K^b(\mathcal{S}^{\text{fin}})$.

Uses Rouquier complexes $\mathcal{R}_i := \underline{B}_i \xrightarrow{\text{blue dot}} R$ ($\mathcal{R}_i$ categorifies T_i)

$$\text{Ev}_{r,s}: \begin{cases} B_i \mapsto B_i & (i \neq n) \\ B_\rho \mapsto R_{n-1} \otimes \cdots \otimes R_1[r] \langle s \rangle \end{cases}$$

- Rouquier calculus only partially developed, but it suffices!

Theorem (Mackaay–Miemietz–Vaz, 2024)

For each (r, s) there is a monoidal functor

$$\mathrm{Ev}_{r,s} : \widehat{\mathcal{S}}_n^{\mathrm{ext}} \rightarrow K^b(\mathcal{S}_n)$$

categorifying the classical evaluation homomorphism.

What this gives:

- A family of triangulated 2-representations of $\widehat{\mathcal{S}}_n^{\mathrm{ext}}$.

Why it matters:

- These are the first evaluation 2-representations in affine type A.

We now do for parabolic induction what we did for evaluation...

- Categorical embeddings

$$\psi_L: \widehat{\mathcal{S}}_k^{\text{ext}} \rightarrow K^b(\widehat{\mathcal{S}}_n^{\text{ext}}) \quad \text{and} \quad \psi_R: \widehat{\mathcal{S}}_{n-k}^{\text{ext}} \rightarrow K^b(\widehat{\mathcal{S}}_n^{\text{ext}})$$

$$\psi_L(B_i) = B_i$$

$$\psi_R(B_j) = B_{j+k}$$

$$\psi_L(B_\rho) = B_\rho R_{n-1} R_{n-2} \cdots R_k$$

$$\psi_R(B_\rho) = R_k^{-1} \cdots R_2^{-1} R_1^{-1} B_\rho$$

- Symmetric braiding $\psi_L \psi_R \cong \psi_R \psi_L$

Theorem (Mackaay–Miemietz–Vaz, 2025)

There is a monoidal functor

$$\Psi_{k,n-k} : \widehat{\mathcal{S}}_k^{\text{ext}} \boxtimes \widehat{\mathcal{S}}_{n-k}^{\text{ext}} \rightarrow K^b(\widehat{\mathcal{S}}_n^{\text{ext}})$$

categorifying parabolic induction.

What this gives:

- A categorified Zelevinsky tensor product of 2-representations.

Why it matters:

- It allows us to build new affine 2-representations from smaller ones.

Example ($n = 2, k = 1$)

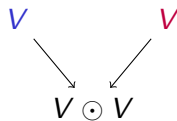
- Let V be the **trivial birepresentation** of $\widehat{\mathcal{S}}_1^{\text{ext}} = \langle B_\rho \rangle$.
- Algebra object for V :** $X \in \widehat{\mathcal{S}}_1^{\text{ext}, \diamond}$ is given by

$$X := \coprod_{r \in \mathbb{Z}} B_\rho^r.$$

Note: $B_\rho X \cong X$.

- Induction:** $Y := \psi_{1,1}(X \boxtimes X) \in K^b(\widehat{\mathcal{S}}_2^{\text{ext}, \diamond})$, where

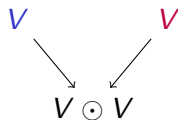
$$Y \cong \coprod_{r \in \mathbb{Z}} \coprod_{s \in \mathbb{Z}} (B_\rho T_1)^r (T_1^{-1} B_\rho)^s.$$



Example ($n = 2, k = 1$)

- Let V be the **trivial birepresentation** of $\widehat{\mathcal{S}}_1^{\text{ext}} = \langle B_\rho \rangle$.
- Algebra object for V :** $X \in \widehat{\mathcal{S}}_1^{\text{ext}, \diamond}$ is given by

$$X := \coprod_{r \in \mathbb{Z}} B_\rho^r.$$



Note: $B_\rho X \cong X$.

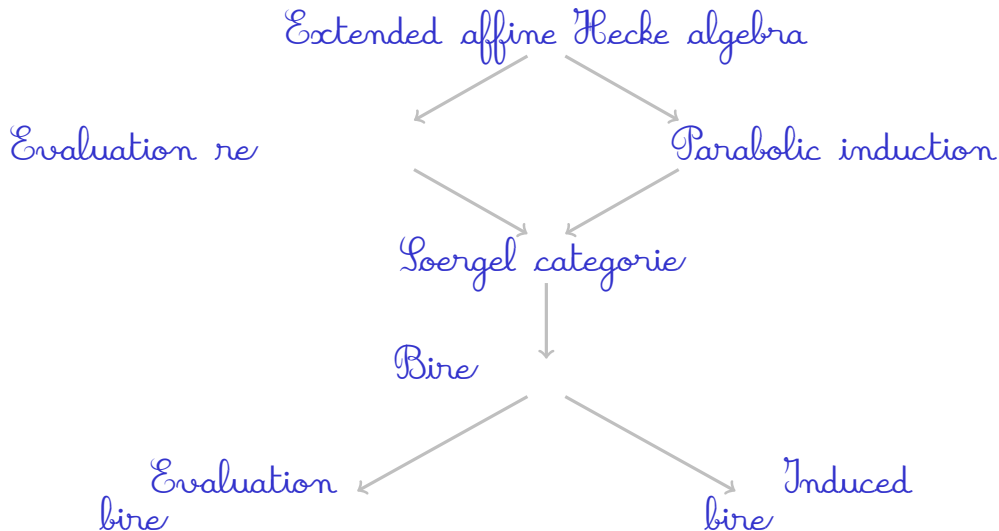
- Induction:** $Y := \psi_{1,1}(X \boxtimes X) \in K^b(\widehat{\mathcal{S}}_2^{\text{ext}, \diamond})$, where

$$Y \cong \coprod_{r \in \mathbb{Z}} \coprod_{s \in \mathbb{Z}} (B_\rho T_1)^r (T_1^{-1} B_\rho)^s.$$

- Induced triangulated birepresentation:** By a general construction due to Fan–Keller–Qiu, there is a triangulated closure of

$$\text{add} \left\{ FY \mid F \in K^b(\widehat{\mathcal{S}}_n^{\text{ext}}) \right\} \subset K^b(\widehat{\mathcal{S}}_2^{\text{ext}, \diamond}).$$

Summary: what did we achieve?



- Theory of irreducibles in triangulated 2-representations.
- How can triangulated 2-reps of affine Soergel categories be classified?
- Is there a triangulated version of the Zelevinsky classification?
- Lift evaluation functor to a functor $\mathrm{Ev}_{r,s}^{K^b}: K^b(\widehat{\mathcal{S}}_n^{\mathrm{ext}}) \rightarrow K^b(\mathcal{S}^{\mathrm{fin}})$.
- Extension beyond affine type A .
- Connections with affine Springer theory, character sheaves, and knot homology.

Thank you!

A diagram in Rouquier–Soergel calculus

